

Feasibility Study

Urban Imageability and Geographic Knowledge in Vision-Language Models · Middlesex University London · Experiment 1 (feasibility) · July 2026

Verdict. The proposed comprehensive design (50 cities, ~200 locations each, ~10,000 locations, 6+ models, city and neighbourhood evaluation) is **feasible** under some conditions. Money is not the main limit (a complete run costs roughly £150 to £600); the main limit is **inference time**. Open-weight models on Colab's free GPU process about one location every 30 seconds, so a full 10,000-location pass takes 3 to 4 GPU-days per model, which on the free tier spreads over 1 to 2 calendar weeks per model. The design I recommend in section 4 keeps the full 50-city breadth and sets locations, input format and the model panel to values that finish in a realistic timeframe. All estimates below come from a 124-location pilot I already ran through four model families.

1 · Reference scenario and the question

The scenario under test is the currently proposed design:

- 50 cities, 8 world regions, stratified random sampling (already drawn: 17,566 points).
- ~200 locations per city, ~10,000 locations for the city task.
- Several image configurations per location (single, opposite pair, four cardinal, 2x2 grid).
- At least six vision-language models, proprietary and open-weight.
- City-level and, where relevant, neighbourhood-level evaluation.

What I test here is whether this design is *executable* within the time, budget and hardware of a student project. I go through each requirement in turn and close with one recommended design.

2 · The end-to-end pipeline (9 steps)

Every stage below is already implemented and exercised on the pilot; the study runs the same code at scale.

#	Step	How it is done	Status
1	Select city and sampling locations	Stratified random draw over the GHSL urban frame, fixed seed, 5 fabric strata	Done (17,566 pts)
2	Query Google Street View	Static API, pano-pinned, 4 cardinal views (fov 90, 640x640); metadata snap within 50 m	Working
3	Download, validate, store	Per-point commit; pano_id, capture date, snap distance logged; dedup by pano	Working
4	Prepare visual input	Text-blur (CRAFT + Gaussian); compose single / pair / four / 2x2 grid	Working
5	Submit input and prompt to each model	Versioned prompt + JSON schema; no tools; open models on Colab, proprietary via API	Working
6	Parse the model response	Structured JSON; tolerant recovery for malformed output (unquoted keys, truncation)	Working
7	Convert prediction to coordinates	Model returns lat/lon directly; free-text city name resolved at scoring	Working
8	Country / city accuracy and localisation error	Geodesic error, accuracy at 25/200/750 km, alias-aware country and region match	Working

9	Store full result and metadata	Raw answers immutable; scores versioned separately; run config recorded	Working
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The whole pipeline is in place. The only stage I have not yet run at full scale is step 5, across the whole corpus and the full model panel.

3 • Estimates (measured from the pilot)

3a • Execution time

Per-location latency, measured on the 124-point pilot:

Model	Environment	sec / location	10,000 locations, serial
SmolVLM-2B	Colab free T4	5.4	15 h
Gemini-3.5-flash	Google API	13.2	37 h (or hours via Batch)
Idefics2-8B	Colab free T4	28.2	78 h
Qwen2.5-VL-7B	Colab free T4	32.9	91 h

A representative 7-8B open model needs about **3 to 4 GPU-days per full 10,000-location pass**. The free Colab tier gives roughly 3 to 5 usable GPU-hours before a disconnect, so in calendar terms each open model takes **1 to 2 weeks**. Proprietary models are faster and, via the asynchronous Batch API, return a full pass within 24 hours. This is the main bottleneck, so I sized the design in section 4 around it.

3b • Monetary cost

Item	Cost
Open-weight models (any number), Colab free T4	£0
Remaining imagery for 10,000 locations (~7,550 new points, 30,200 views, £5.25/1,000)	£159 (or £0 paced)
<i>Proprietary models, economical tier (full study, Batch):</i>	
Gemini Flash	£40–80
Claude Haiku 4.5	£36
GPT-mini	~£15–30
Top-tier alternatives (optional, upper bound): Claude Sonnet 5 / GPT flagship	£107 / ~£350
Google Cloud research credits available (to 29 Sep)	–£223

All proprietary figures are compared on the economical tier of each provider, since the pilot showed this class is already accurate enough; the top-tier row is only an upper bound on spend. GPT-mini has not been run in the pilot, so its figure is an estimate from public pricing, not measured. Pilot actual spend: £6.58 for 248 Gemini requests, which calibrates the API rates. Open-weight inference has no marginal cost.

3c • Data-acquisition scalability

Street View collection is the slowest external dependency. The free tier covers about 10,000 image requests per month, i.e. ~2,500 locations per month at 4 views each. 2,450 locations are already in hand. The remaining corpus therefore takes either ~£159–317 to buy immediately (about 2 weeks) or several months at £0 on the free tier. Storage measured from the pilot is 788 MB of originals per 2,450 locations; with text-blurred copies that is about 0.65 MB per location, so 10,000 locations need roughly **6.5 GB** and the full 17,566-point corpus about 11 GB, both of which fit on a laptop disk.

3d • Input-configuration comparison

The four configurations differ mainly in the number of image tokens, which drives both cost and speed:

Configuration	Images sent	Relative token load	Notes
Single view	1	lowest	Cheapest, fastest; least context
Two opposite views	2	~2x	Some context, moderate cost
Four cardinal (separate)	4	highest	Full context; excludes single-image models
2x2 grid (composite)	1	low	Full context in one image; works on every model

On feasibility grounds I would default to the **2x2 grid**: it carries all four views in a single image, keeps the token load close to the single-view level, and runs on every model including single-image ones. Whether the extra context actually improves accuracy over a single view is an empirical question, and Experiment 2 answers it (same 124 points, 12 cities, 3 models, all four configurations); the infrastructure to run it exists and two of the four configurations are already measured.

4 • Recommended experimental design

A design that preserves the full 50-city, 8-region breadth and the multi-model comparison, while finishing in a realistic timeframe on free hardware.

Number of cities	50 (the full sampled frame, 8 world regions)
Locations per city	~100 for the city task (5,000 total), balanced across the 5 fabric strata; extend to 200 later if time allows
Images per location	1 composite 2x2 grid of the four cardinal views (text-blurred)
Models	6: open, Qwen2.5-VL-7B, Idefics2-8B, InternVL or MiniCPM-V, plus one more open; proprietary, Gemini-flash and Claude Sonnet 5 (GPT optional)
Input format	Single multi-image prompt, structured JSON output, no tools
Estimated inference calls	5,000 locations x 6 models = 30,000 (city task); +neighbourhood task where eligible
Expected execution time	Open models ~1 week each on free Colab (or ~2 days on Colab Pro); proprietary ~1 day each via Batch; ~3 to 4 weeks calendar with overlap
Expected monetary cost	~£50 to £150 (imagery top-up + proprietary Batch, less the £223 credits); open models £0
Storage requirement	~3.2 GB for 5,000 locations (originals + blurred)
Necessary hardware	Laptop + database; Google Colab free T4 (Colab Pro at ~£10/month optional, to remove the time bottleneck)
Main technical risks	Colab session limits (serial inference is slow); text-blur recall; proprietary API cost drift
Contingency plan	If runtime or cost exceed estimate: drop to 50 locations/city or fewer models, switch open runs to Colab Pro or university HPC, and restrict proprietary models to Batch mode; every run is resumable, so partial progress is never lost

5 • Coverage of the four experiments

Experiment	Status
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1 · Feasibility	This report; pipeline documented, estimates measured, design recommended
2 · Effect of visual context	Infrastructure ready; 2 of 4 configurations measured; full comparison is the next run
3 · Internal knowledge only (no retrieval)	Open models have no retrieval mechanism at all; proprietary calls pass no grounding tool and record <i>retrieval_disabled</i> ; to be documented formally with prompt and environment
4 · Which models are suitable	Pilot shows Gemini and Idefics2 locate well, Qwen weakly, SmolVLM-2B effectively guesses (country 10%); minimum capability is around the 7-8B open class

6 · What is already built (why the risk is low)

- Sampling frame complete: 17,566 Street-View-verified points, 50 cities, 545 eligible neighbourhoods.
- Database (PostgreSQL/PostGIS), ingest pipeline and inference harness built and unit-tested (105 tests).
- Colab multi-model worker validated on four model families; scoring and mental-map analysis working.
- 2,450 locations of imagery already collected and text-blurred.